TEACHER GUIDE

AQA GCSE Business (8132)

Financial calculations: ARR



3.6 Finance

3.6.3 Financial terms and calculations: average rate of return

A revision lesson. Pupils run a car dealership in firms of four and revise financial calculations: ARR on their own deals. Seat each firm together. Each firm answers the topic tasks together.

The lesson

45-minute setting

Setup	Round 1	Round 2	Round 3	Final questions

EVERY ROUND

Firm decisions
3 min

→

Deals
4½–5 min

→

Topic tasks
4 min

→

B2B deal
2½ min
from Round 2

LESSON SETTINGS

30 min	45 min	60 min	75 min

What pupils do

Every pupil buys and sells cars. In Firm decisions, each role has its own job.



CEO

Staff · Sabotage from Round 2



Finance

Capital & Loans · Investments



Sales & Marketing

Marketing



Operations

Seasonal stock

Topic tasks

Each firm gets its own figures. The method is the same. Your Live screen shows each task and every firm's answer.

1 Lifetime profit

Round 1

A new valeting bay.

Total profit over its life?

Cost to set up	**£45,000**
Returns, each year	**£14,500**
It lasts	**5 years**

ANSWER

£27,500

5 × £14,500 − £45,000 = £27,500

Why: It earns back its cost and £27,500 more.

2 What ARR misses

Round 1

You compare two investments by ARR.

Which two things does ARR not show?

- **A** The cost of the investment
- **B** When the money comes back
- **C** The average profit a year
- **D** The risk that returns fall short
- **E** Profit as a % of the cost

ANSWER

✓ When the money comes back
✓ The risk that returns fall short

Why: ARR averages the profit; it says nothing of timing or risk.

3 ARR steps in order

Round 2

You work out the ARR of a new ramp.

In what order do you work it out?

- **A** Divide by the years
- **B** Add up all the returns
- **C** Divide by the cost, × 100
- **D** Take off the cost: total profit

ANSWER

1. Add up all the returns
2. Take off the cost: total profit
3. Divide by the years
4. Divide by the cost, × 100

Why: Total profit, then profit a year, then as a % of the cost.

Topic tasks

4 Valeting bay ARR

Round 2

A new valeting bay.

Average rate of return (ARR), as a %?

Cost to set up	**£60,000**
Profit over its life	**£84,000**
It lasts	**5 years**

ANSWER

28%

£84,000 ÷ 5 = £16,800; ÷ £60,000 × 100 = 28%

Why: It earns 28p a year for every £1 put in.

5 ARR below loan rate

Round 3

A ramp would have an ARR of 6%. You would borrow the money at 8% a year.

Should you invest, and why?

A No

B Yes

ANSWER

✔ **No**

Why: A 6% return does not cover an 8% cost of borrowing.

6 Better option ARR

Round 3

Which one pays better?

ARR of the better option, as a %?

Option A: cost	**£75,000**
Option A: profit each year	**£9,000**
Option B: cost	**£55,000**
Option B: profit each year	**£13,750**

ANSWER

25%

A 12%; B 25%; better 25%

Why: Compare the rates, not the pounds of profit.

Final questions

Each pupil answers 6 final questions on financial calculations: ARR. They come from these 10 questions.

Then each pupil writes one short answer: “Explain why ARR is compared with the interest rate.”

1 A workshop upgrade costs £60,000 and is forecast to make total profit of £15,000 over three years. What is its average rate of return?
✔ **8.3%**

2 A project is forecast to make £3,000, £5,000 and £4,000 profit over three years. What maximum investment gives a 20% average rate of return?
✔ **£20,000**

3 Upgrade A: cost £50,000, £12,000 profit over three years. Upgrade B: cost £40,000, £10,800 over three years. Which has the higher average rate of return, and by how much?
✔ **Upgrade B, by 1 percentage point**

4 Two investments have similar average rate of return. What additional factor should the Finance lead consider?
✔ **Risk and when the cash comes back**

5 A new MOT bay costs £40,000 and should make £24,000 total profit over four years. What is the average rate of return?
✔ **15%**

6 A paint booth costs £80,000 and is expected to make £10,000 profit every year. What is its average rate of return?
✔ **12.5%**

7 Project A costs £30,000 and makes £9,000 total profit over three years. Project B costs £50,000 and makes £12,500 over five years. Which has the higher average rate of return?
✔ **Project A, at 10%**

8 What does the average rate of return tell a dealer about an investment?
✔ **Average yearly profit as a percentage of the cost**

9 A dealer spends £40,000 on a car wash. Over 4 years it makes a total profit of £24,000, after taking off the cost. What is the ARR?
✔ **15%**

10 A dealer can buy a car wash with an ARR of 12% or a new ramp with an ARR of 8%. On ARR alone, which should it choose?